

Identity, Not Attributes: What Configuration Data Contributes to Incident Prediction in Two Organisations

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Abstract

A configuration management database (CMDB) programme is justified by the analytics it enables, and that justification is a comparison against a baseline the analyst chooses. We measure what configuration data contributes on a named operational task in two organisations, and find the answer set less by the data than by two decisions that are rarely written down: which layer of the configuration hierarchy is being bought, and which already-recorded fields the comparison baseline is allowed to contain.

On a public event log of 45,455 incidents from a bank’s IT operation, predicting at intake whether an incident will be reassigned, a 256-way service-component grouping captures three quarters of what instance-level identity is worth, and instance identity adds +0.023 AUC over it. A per-item outcome rate — no model, and no configuration attribute of any kind — scores 0.744 against the full model’s 0.748. What the CMDB supplies on this task is a stable identifier under which outcome history accumulates, not the configuration attributes it is bought for.

The second decision explains why the headline number moves. Knowing the affected item is worth +0.183 AUC against four intake fields and +0.103 [+0.094, +0.113] once one further field the organisation already records is admitted: which group logged the incident. Across every design choice we varied — split point, target threshold, cleaning cutoff and estimator family — the reduction ranges 36.1% to 48.3%, and the ordering never reverses. Four free creation-time congestion features leave it there. The pattern replicates on a second organisation, tool and country: on the BPI Challenge 2013 log

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the same ladder gives a reduction of 61.3% [54, 68] on that challenge’s own ping-pong target and 43.9% [31, 55] at a stricter threshold, though the free field is more tightly coupled to the target there, so we read it as an upper bound.

We report what the measurement does not support in the same detail. Eight errors of our own are reported as results rather than edited away, two of them found in the round that produced this version. The larger of the two is this paper’s own: an operational overstatement we had reported as a factor of 4.3 at a fixed review capacity does not survive, because its sign reverses when ties inside the naive baseline’s scores are broken differently. Restated as net benefit across the whole range of decision thresholds, omitting the free field overstates the item’s value by 1.07 at the threshold where that value is largest.

Keywords: configuration management database, predictive process monitoring, baseline specification, variable importance, IT service management, incident management, decision curve analysis

1. Introduction

A CMDB programme is expensive and is justified by what it will enable. That justification rests on a comparison whose second arm — performance without configuration data — is a choice the analyst makes, usually silently, by deciding which already-available fields to include. It also rests on a quantity that is almost never separated out: which *layer* of the configuration hierarchy the benefit comes from, given that an instance-level register costs far more to build and maintain than a service register.

This paper measures both, on two organisations’ incident logs, and reports three things.

1. **Identity, not attributes, and not at instance level.** A 256-way service-component grouping of the same estate captures three quarters of what instance-level identity is worth on this task, and instance identity adds +0.023 over it. Below that, a per-item outcome rate applied as a lookup table — no model, no other field, no configuration attribute — reaches 0.744 against the full model’s 0.748. What the configuration data supplies here is a stable key under which six months of outcomes accumulate (Section 5).

2. **The measured value turns on a field-admission choice.** Against four intake fields, item identity is worth +0.183 AUC; against those fields plus one the organisation already records for nothing, +0.103 [+0.094, +0.113]. The reduction runs 36.1% to 48.3% across the design space and replicates on a second organisation, tool and country (Section 6).
3. **Eight corrections, reported as results.** Six were found in earlier rounds; two in the round that produced this version. Both of the new ones removed a claim the previous version made, and one of them is the paper’s own operational headline (Section 11).

What kind of contribution this is.. We report no deployed system. This is a retrospective study on two public benchmarks, offered as an evaluation practice. Section 8 states what the choice costs in units closer to those a service desk operates in, and Section 5 states which *layer* of configuration data carries the value, which is the question a buyer actually faces.

On what this paper used to claim.. Earlier versions reported larger findings withdrawn under adversarial review, each an artifact of a control we built ourselves. Section 11 lists eight corrections that survive here as reported results rather than silent edits, because each is the kind of error this paper is about. Two were found in the round that produced this version, and one of them — a claim about a public dataset that our own repository contradicted — had stood through eight rounds of review.

2. Background

Prediction on incident event logs.. Our task sits at the boundary of predictive process monitoring, surveyed and benchmarked across real-world logs [24, 25] on data of the kind process mining formalises [1]. We predict at $t = 0$ from creation-time attributes with no prefix, which is tabular classification on a process log rather than prefix-based monitoring in the sense those benchmarks evaluate; we note that Teinemaa et al. [24] explicitly excluded the 2013 and 2014 BPI Challenge logs from their benchmark for lacking a static/dynamic attribute split. The BPI Challenge logs are the standard public instances [8, 22]; comparable exports exist elsewhere [3]. Such logs carry documented quality hazards [23]. One hazard bears directly on us and we state it as our own measurement rather than attributing it: values recorded once per record can be replicated onto every event of that record,

which makes constancy across a case uninformative about when the value became known. Establishing what a resource stamp on an event denotes is the business of organisational mining [2], which is what Section 4 does from first principles for the field our result turns on. The state of the queue a case arrives into — the inter-case perspective — carries real predictive content in this setting [21], and Section 6 admits four free creation-time congestion features to the baseline for that reason.

The CMDB and what justifies it.. A CMDB records configuration items and their relationships, and is the asset on which the configuration-management practice of ITIL and its successors rests. Survey work on ITSM framework adoption [15] finds that realised benefits grow with implementation maturity, but measures them as practitioner-reported outcomes rather than as the analytic value of the configuration data itself.

What is and is not already measured.. Prior peer-reviewed work uses the affected configuration item as one input among many without isolating its value. Schad et al. [20] predict ticket reassignment — our own target — with a relational graph convolutional network on a public ServiceNow log, with the configuration item among the incident-node features. Sarnovský and Surma [19], working on the same Rabobank log we use, report ROC AUC and rank CI type, CI subtype and service component among the most significant attributes. Kapel et al. [11] predict change-induced incidents at a large bank and rank CI Name third by SHAP importance, behind two free-text fields. None of the three performs a controlled with-and-without comparison attributing a change in a fixed metric to configuration-item identity against a stated baseline, which is what we do; but the reader should note that the question is not untouched, and that in the one study which ranks a free-text field against the configuration item, free text wins.

The methods a CMDB is bought to enable.. Surveys of AIOps for failure management [30, 17, 16] catalogue triage, failure-prediction and root-cause methods that take configuration data as input. Incident triage in particular — deciding which team owns an incident — has been treated as a learning problem on operational logs at industrial scale [5], as has failure prediction across a cloud estate [14].

What the comparison is measured against.. The value of a feature *given* a baseline is what algorithm-agnostic variable-importance frameworks formalise; Williamson et al. [28] develop this for R^2 -type predictiveness and

Williamson et al. [29] extend it to AUC, which is our metric. Our overlap is the redundancy that additive importance measures share credit for [7]. Section 3 says what this paper adds to those frameworks and what it does not.

The estimand itself — the change in a fixed metric when one variable joins an existing model — has been argued over in biostatistics for two decades, and that literature reached our conclusion first. Cook [6] shows that an increment in AUC is a poor measure of a variable’s incremental value and depends heavily on the model it is added to. Vickers and Elkin [26] supply the principled replacement for the capacity-limited framing we used in earlier versions and withdraw in Section 8. We claim no priority over these; we report that the failure mode they describe is, on operational data, large enough to change a procurement decision.

Within machine learning, Rendle et al. [18] and Ferrari Dacrema et al. [9] show insufficiently tuned baselines accounting for reported gains in recommender systems; benchmark conclusions are sensitive to variation that is seldom reported [4], and documentation of the choices producing a result is thin [10]. Where the choice concerns which fields may enter, it shades into leakage [13, 12], whose formal treatments accommodate admission rules as constraints set by the problem definition rather than derived from it. In predictive process monitoring, Weytjens and De Weerd [27] show that avoiding a different leak — case prefixes shared between training and test under naive temporal splitting — needs explicit construction; we adopt their strict temporal splitting and, at their prompting, examine the extract boundary in Section 4. We report a field-admission failure mode for a deployment decision rather than a leaderboard, where it is not a modelling detail but the number the business case turns on.

3. This Is Not Just Conditional Variable Importance

The objection we expect first, and the one this paper would deserve if it stopped at Section 6, is that measured importance is baseline-relative, that Williamson et al. [29] and Covert et al. [7] formalise exactly that, and that a study reporting it on one log is reporting a known fact with a new number attached. The objection is correct about the formal content and we concede it in those terms. Four things are added here that no formal framework supplies.

A magnitude on a named decision.. Those frameworks establish that a feature’s importance is defined only relative to a baseline. They do not say how large the dependence is for any particular feature on any particular decision, and a procurement decision turns on the size. On this task the answer is that the choice of baseline moves the measured value of an entire data programme by roughly half, and that half is the difference between an investment case and none.

The right estimand is not an average over baselines.. A Shapley-style attribution averages a feature’s contribution over subsets of the others, which is the correct thing to compute when the question is how to apportion credit within a fixed model. It is the wrong thing to compute here. A business case needs a feature’s value against *the baseline the organisation will actually have* — the fields already in its ticketing tool — not an average over baselines it will never be in. We therefore report a ladder over named, admissible baselines rather than a single attribution, and Section 4 states the admissibility criterion before applying it.

The overlapping field is free.. The interesting case is not that some baseline choice changes the answer; any correlated feature does that. It is that the field which absorbs half the measured value is one the organisation *already records at no cost*, sitting in the same export, requiring no programme and no budget line. That is what makes the analyst’s choice invisible rather than merely arbitrary: nothing in the data flags the omission, and the omitted field has no price tag to prompt a question about it.

It replicates across organisations, and no framework predicts that.. Baseline-relativity is a theorem; whether a particular pair of fields overlaps in a particular estate is an empirical question with no general answer. That the same ladder produces the same ordering and a reduction of the same order on a second organisation, a second tool and a second country (Section 6) is evidence about how ITSM data is generated, not about the definition of variable importance.

What we do not claim.. We do not claim a new estimator, a new importance measure, or priority over Cook [6], who stated the underlying point for AUC in 2007. The contribution is measurement and evaluation practice, and Section 5 rather than Section 6 is where the paper’s novel finding sits.

4. Data and Task

4.1. The primary organisation

We use the incident records of the BPI Challenge 2014 log [8] from Rabobank Group ICT, recorded in HP Service Manager, with the activity log shipped alongside. The configuration-item column we use is `CI Name (aff)`, the item affected at the time the incident was recorded, not the closure-time `CI Name (CBy)`.

Cleaning. Of 46,809 rows, 203 carry no incident identifier and one an unparseable timestamp. A further 1,150 opened before 2013-10-01 are left-censored — already open when the extract began — and are reassigned at 81.2% against 40.0% for the incidents we keep. Removing them leaves **45,455** incidents to 2014-03-31; a temporal 70/30 split gives 31,818 training and 13,637 test, positive rates 0.413 and 0.372. The cutoff is not taken from the outcome: monthly volume rises from 857 incidents in September 2013 to 8,606 in October. The extract is right-censored too — incidents opened near its end cannot have long lives — so we re-measured on cohorts truncated up to a month early; the shrinkage moves between 42% and 45%, so the boundary does not drive it. The affected-item field is 100% populated across 2,929 items, 2,554 seen in training. That population rate is a property of the extract and of a tool configuration that copies the item onto the incident from the originating interaction record; it is not evidence about CMDB accuracy in general, and Section 10 returns to the point.

Task and estimator. Predict at creation whether an incident will later be reassigned. We call the target reassignment rather than misrouting throughout: it fires on 37.2% of test incidents, which is routine handling and not error, and Section 8 reports the ladder at stricter thresholds. Median handling time rises from 1.59 hours for incidents never reassigned to 47.53 hours for those reassigned five or more times; we use this to motivate the task, not as a recoverable saving.

The intake block is four fields carried on the incident form: `Category`, `Impact`, `Urgency` and `Priority`. It is effectively three. `Priority` is a deterministic function of (`Impact`, `Urgency`) on this log — 19 occupied cells, none carrying more than one `Priority`, across all 46,809 rows — so it adds no information, and dropping it moves intake AUC from 0.562 to 0.564. Sarnovský and Surma [19] reported this redundancy on this same log in 2018. We keep the column because it is what an analyst building the naive arm would

include, and we disclose it because a paper arguing that baselines are under-specified should not ship an under-specified baseline in silence. The four fields together take only 23 distinct combinations in training and 23 in test, which is a fact Section 8 needs and which no choice of estimator can change.

The primary estimator is one-hot logistic regression, chosen because it is bin-free: scikit-learn’s `LogisticRegression` with L2 penalty, inverse regularisation strength $C = 1.0$, the `lbfgs` solver and a cap of 3000 iterations, with `OneHotEncoder` set to ignore unseen levels. The penalty is held fixed across every arm; tuning it per arm on an inner split of training data moves the headline reduction from 43.7% to 43.2%. Two further estimator specifications are reported in Section 6. Encoders and rankings are fit on training data only; intervals are 2,000-resample paired bootstraps, except Table 3, whose counts are re-ranked in each draw and use 400. Monte-Carlo nulls use 15 to 100 draws, the count set by cost and recorded in the code that produces each. Every sampling script is seeded from a single constant, so the reported figures are reproducible exactly.

The free field, and what it is not.. The activity log records, on the `Open` event of every incident, an assignment group: 49 in training, none missing. An earlier draft of this paper called it the queue the service desk routed the incident to. That was an assertion, not a finding, and it is wrong. Three checks say so. The `Open` rows carry 50 distinct groups where the `Assignment` rows carry 218, and a routing destination cannot be less diverse than the teams that then do the work. One group accounts for 67.0% of `Open` rows but only 18.4% of all activity rows. And the `Open` group matches the group on the incident’s first `Assignment` activity for just 15.1% of the 37,577 incidents that have one. The field is the group that *logged* the incident: one central desk for two thirds of tickets, a long tail for the rest. We do not characterise that tail — its openers match the first `Assignment` group only 21.8% of the time and appear anywhere in the later work rows only 59.1%, so calling them teams opening their own work would be another assertion of the kind we are correcting. It is also concentrated and unstable: the largest group holds 62.1% of training incidents, and the number of live groups falls from 49 to 32 between our two halves, while in the test half it logs 78.6% of incidents.

The free field is not a restatement of the target.. A reader may reasonably suspect the opposite of what Section 7 argues: that a field naming a *group* cannot help but encode a target defined as a *change of group*, making the

whole ladder circular. That reading makes a directional prediction, and the data contradicts it. The central desk logs most tickets and does a small minority of the work, so on the circular reading it should be the one handing work on, and should reassign more than everyone else. It reassigns *less*: 0.309 against 0.603, a difference of -0.294 $[-0.315, -0.275]$. What the field separates is a low-reassignment intake channel from a high-reassignment one. Nor is it a strong predictor in its own right: the per-group outcome rate, applied as a lookup with no model, scores 0.642, and the one-bit central-desk contrast 0.606. A field that restated the target would score near the ceiling.

Why not the real routing decision.. The obvious repair is not available. The first **Assignment** group occurs strictly after **Open** on every incident that has one, a median of 46 minutes later, and 7,878 of 45,455 never have one. It fails the criterion below, and substituting it would introduce the leakage that criterion prevents.

Admissibility, and mutation.. A field on the **Open** row is admitted only if it is a per-event observation, tested by whether it varies across an incident's activity rows; the opening group varies for 92.56% of cohort incidents and matches the last-observed group for only 21.35%. Because the incident file is a closed-record snapshot we also report how often each field is edited after creation: Urgency on 1,107 incidents (2.44%), Impact on 1,084 (2.38%), the affected item on 159 (0.35%). Incidents whose Impact was edited are reassigned at 0.66 and those whose Urgency was edited at 0.67, against 0.39 for the rest, so this is not neutral; Section 6 reports the headline with those fields dropped and with the edited incidents removed. The activity log carries no Category or Priority change type, so for those two fields the snapshot cannot be audited at all; our disclosure covers the fields for which an audit exists, not every field we use.

A second criterion governs the other direction, and Section 5 needs it: a field may join the CMDB-free baseline only if the organisation would hold it *without* the CMDB. That excludes every field which is a function of item identity, because such a field exists only where the item register does.

4.2. The second organisation

Until this round we described this study as single-organisation of necessity, on the ground that of the three public ITSM incident logs in common use, one records the affected item on every incident, the second on 0.2%, and the third

had no such field at all. The third clause was false, and our own repository contained the evidence: the BPI Challenge 2013 incident log [22] from Volvo IT, recorded in VINST, carries a **product** attribute with 704 distinct values, exactly one per trace, on all 7,554 traces and none missing. It is a coarser identifier than an instance-level CI name, but it is an affected-item field, and the challenge’s own headline question was ping-pong behaviour between teams — the same outcome our primary target measures.

We therefore run the ladder there too. The item analogue is **product**; the free field is the **org:group** on the first event, of which there are 338; the intake block is **impact**, the one intake-form field the two logs share. The target is at least one change of **org:group** along the case, which fires on 50.6% of cases, with a stricter variant at two or more changes (27.2%). We use the same temporal 70/30 split and the same estimator.

One difference cuts against us and we state it rather than let a reader find it. On BPI Challenge 2014 the opening group matches the incident’s first **Assignment** group only 15.1% of the time: it is the desk that logged the ticket, a different object from the routing sequence that defines the target. On BPI Challenge 2013 the opening group *is* the first term of the **org:group** sequence whose variation defines ping-pong. The coupling is therefore strictly tighter there, the free field is correspondingly worth more (+0.188 against +0.082), and the Volvo reduction should be read as an upper bound rather than as a second draw from the same distribution.

5. Which Layer of Configuration Data Pays?

The field-admission question of Section 6 has a companion that is more directly a buying decision, and it is where this paper’s novel finding sits: not whether configuration data is worth anything, but at what resolution, and whether what pays is the configuration *attributes* at all. Four nested groupings of the same estate are all 100% populated and free once the item is known. Measured over the intake-plus-group baseline of 0.644:

A 256-way service-component grouping captures three quarters of what instance-level identity is worth, and instance identity adds +0.023 over it. The classification layers proper — CI type and subtype — carry little, and CI Subtype scores below CI Type despite being finer, which is what a fixed penalty does to a sparser column. This is the same +0.023 that Section 7 reports for admitting the service component to the baseline, seen from the other side: the two are the same quantity, and a reader who regards an application

Grouping	levels	AUC	gain
CI Type (aff)	13	0.671	+0.027
CI Subtype (aff)	61	0.657	+0.013
Service Component WBS (aff)	256	0.722	+0.078
CI Name (aff)	2,554	0.748	+0.103
CI Name, marginal over the service component	—	—	+0.023

Table 1: At what resolution configuration data carries its value, over a baseline that already contains the opening group. Levels are counted in the training half.

or service register as something an organisation has without a configuration-management practice should read the CMDB’s marginal contribution here as that figure.

Why these fields are in the ladder and not in the baseline.. The obvious question is why CI Type and CI Subtype, being 100% populated and free, are not simply admitted to the CMDB-free baseline alongside the opening group. The answer is the second admissibility criterion of Section 4 and it needs no model: both are exact functions of item identity. Neither varies on a single one of the 2,929 items, so both exist only where the item register exists, and admitting them would be admitting the CMDB to the CMDB-free baseline. The service component is the case this argument does not dispose of, and the quantitative gap is instructive: it varies on 58 of 2,929 items, carrying 8.7% of incidents, where the opening group varies on 565 items carrying 92.5%. The line between the two fields is real and it is a matter of degree, which is why we report the headline both ways rather than choosing.

The signal is an outcome history keyed on identity.. A per-item reassignment rate computed on the training half and applied to the test half as a lookup table — no model, no other field, no configuration attribute of any kind — scores 0.744, against 0.745 for item identity alone under the full estimator and 0.748 for the paper’s complete model. Whatever the CMDB supplies here, it is not the item’s attributes; it is a stable identifier under which six months of outcomes can be accumulated. The defensible reading is that maintaining such an identifier across a changing estate — dedupe, reconciliation, retirement — is itself a configuration-management function, and we think that reading is right. But it is a different claim from “the CMDB is

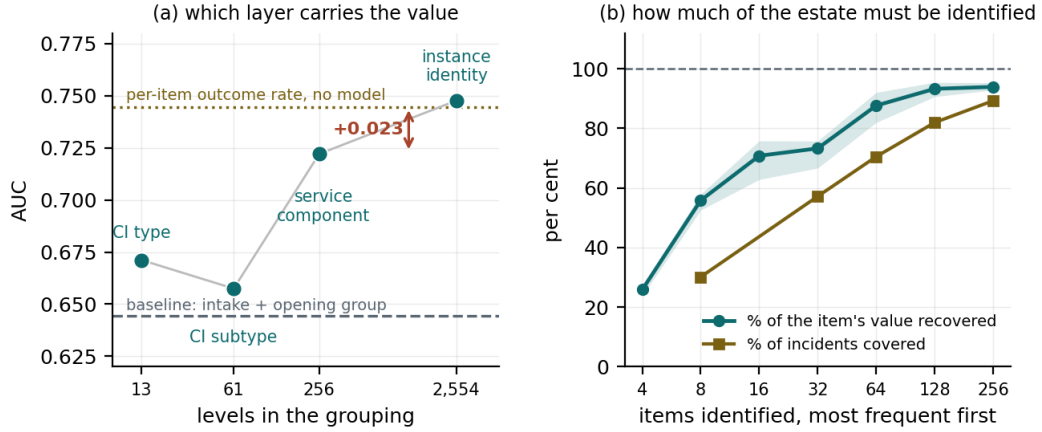


Figure 1: (a) Which layer carries the value, over a baseline that already contains the opening group. The dotted line is a per-item outcome rate applied as a lookup, with no model and no configuration attribute. (b) How much of the estate must be identified: the share of the item’s measured value recovered by identifying only the k most frequent items, against the share of incidents those items account for. Band: min–max across five split points.

worth +0.103”, and an organisation that can key its ticket history on any stable token can have most of this without one.

Estate concentration.. Required scope is bounded by a property of the estate needing no model: the top 8 items generate 30.2% of training incidents, the top 64 70.5%, and the top 128 — 5.0% of the training vocabulary — 82.0%, all from a frequency count before funding anything. Identifying only those items recovers most of the value. Averaged over five split points, the top 8 recover 56% of the +0.103, the top 64 recover 88% and the top 128 recover 93%; across those splits the three range over 52–58%, 81–93% and 90–96%. These are min–max spreads over a design choice, not bootstrap intervals, and we write them as ranges to keep the distinction. We give the mean and range rather than one split’s curve because a single curve is not monotone at this resolution: at $k = 32$ the across-split spread is 9 points. The same holds with the group removed from the baseline — 89% at $k = 64$ — so this is a property of the estate’s concentration, not of our field-admission decision. Taken with the resolution ladder, the practical reading is that the marginal AUC of the 2,801 items outside the top 128 is small, and that is where the cost of a CMDB programme lives.

Organisation	Baseline	AUC	+ item id.	gain [95% CI]
Rabobank (BPIC 2014)	intake only	0.562	0.746	+0.183 [+0.172, +0.195]
	+ opening group	0.644	0.748	+0.103 [+0.094, +0.113]
Volvo IT (BPIC 2013)	intake only	—	—	+0.238
	+ opening group	—	—	+0.092

Table 2: The value of knowing the affected configuration item, with and without a field the organisation already records, on two organisations. The Volvo rungs use a coarser item identifier and a more tightly coupled free field; see Section 4.2.

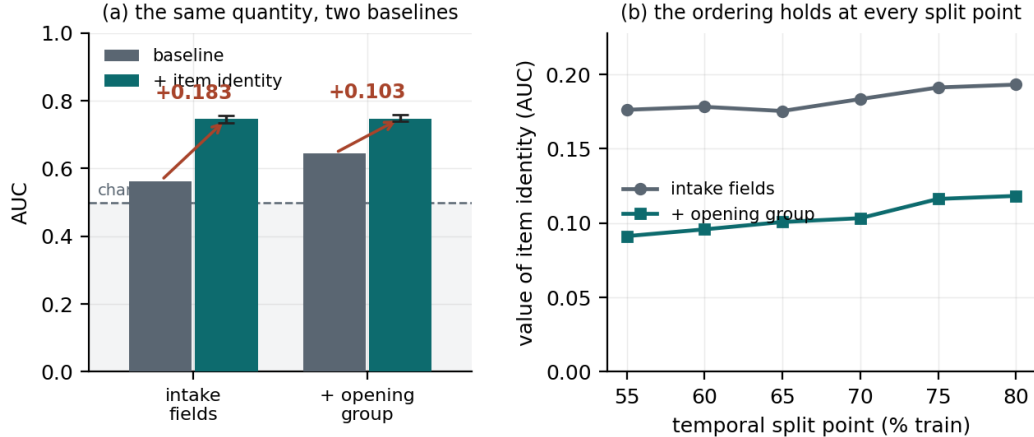


Figure 2: (a) The same quantity under two baselines, on the metric’s full range; the whisker on each treatment bar is the paired bootstrap interval of that bar’s own gain. (b) The ordering holds at every temporal split point from 55% to 80%.

6. The Admissibility Effect

Against four intake fields the affected item looks decisive. Adding the opening group cuts its measured value by 44%. The gains are roughly 28 and 17 standard deviations outside a null in which the incident-to-item association is shuffled, preserving column count and marginal distribution while destroying the signal — we round because re-drawing the same null at a different draw count moves the first figure by about 0.7 — a comparison that matters because adding item identity adds 2,554 indicator columns, which is not free under a fixed penalty. These z figures pool the null’s Monte-Carlo dispersion with the gain’s own bootstrap standard error; dividing by the null alone, as an earlier draft did, would report 51 and 33. Because these nulls

use as few as 15 draws, a two-digit z is an extrapolation from a dispersion estimate and not a tail probability; the load-bearing statement is that no null draw comes near either gain.

It is not a split artifact: across six split points the two gains range +0.175 to +0.194 and +0.091 to +0.119 (Figure 2b), and across split point, target threshold and cleaning cutoff the second ranges +0.067 to +0.130 — wider than any bootstrap interval, which conditions on all of those. Nor does the disclosed field editing explain it: reducing the intake block to Category alone gives +0.106, and restricting to the 44,227 never-edited incidents gives +0.107.

The reduction, over the whole design space.. The reduction rather than either gain is what we argue transfers, so it needs the same disclosure. Across split points it runs 38.7% to 48.3%; across the three target thresholds 43.6% to 48.3%; across the three estimator specifications 42.7% to 47.1%; and across five cleaning cutoffs from no censoring removal at all to a two-month truncation, 36.1% to 44.0%. Over the whole design space it runs **36.1%** to **48.3%**. The bootstrap interval of [40, 48] we report at the published setting conditions on every one of those knobs, and an earlier version of this paper quoted it without saying so while asserting in its title that the reduction was nearly a half. At the low end of the design space it is closer to a third. What survives across the space is that the reduction is large and its sign never changes.

It is not congestion.. A reader from the process-mining community will ask whether the affected item is standing in for load, since the inter-case perspective is known to carry real content on this kind of log [21] and neither of our baselines looks at the state of the queue an incident arrives into. Four features are free at creation and are computed from the open and close timestamps of *other* incidents, every one of which is in the past at the moment of creation: the open backlog, arrivals in the preceding hour, hour of day and day of week. Continuous features are cut at training deciles, so the estimator is unchanged. Admitted to the baseline on the same criterion as the opening group, they move the item’s measured value from +0.103 to +0.100, and the reduction from 43.7% to 45.7% [41, 50]. Alone, without the intake block or the group, the same four features score 0.497 — at this horizon, on this log, creation-time congestion carries essentially nothing. The item is not a load proxy.

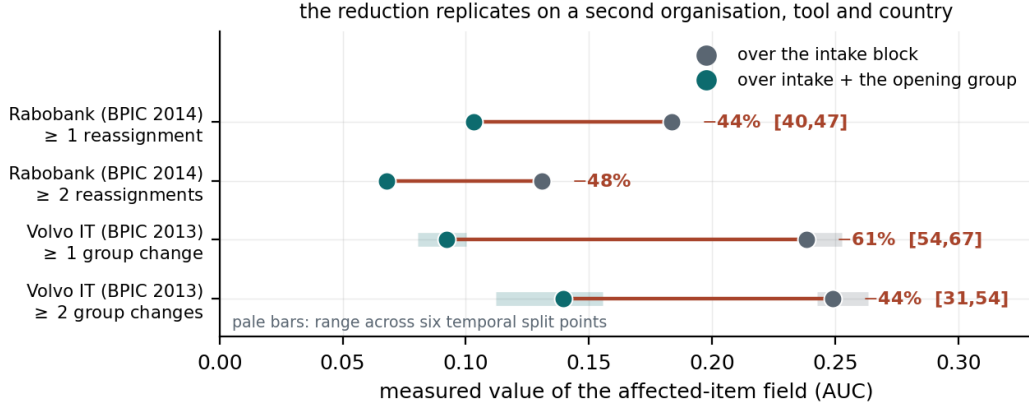


Figure 3: The ladder on both organisations, at two target thresholds each. Each segment runs from the item’s value over the intake block to its value once the opening group is admitted; the label is the reduction, with its paired bootstrap interval where one was computed. Pale bars are the min–max range across six temporal split points, which was computed for the Volvo rungs only.

It replicates on a second organisation.. On Volvo IT the same ladder gives +0.238 against intake and +0.092 once the opening group is admitted, a reduction of 61.3% [54, 68]; at the stricter two-change threshold, +0.249 and +0.139, a reduction of 43.9% [31, 55]. Neither interval includes zero. Across six temporal split points the first runs 60.3% to 67.8% and the second 40.8% to 53.8%. The qualitative claim — that admitting one field the organisation already records removes a large fraction of the item’s measured value — holds on a second organisation, a second tool and a second country. The magnitude does not transfer, and Section 4.2 gives a structural reason to expect the Volvo figure to be the larger of the two.

It is not an artifact of the estimator.. The dimensionality null rules out the regularisation burden of 2,554 sparse columns, but a reader may want the effect shown where that burden cannot arise. We re-represent item identity as a single cross-fitted target-encoded column — fitted on training only, out-of-fold, so no row sees its own outcome — which makes the item one continuous feature and makes histogram boosting usable on a variable that otherwise collapses 2,554 items into 137 bins. Across one-hot logistic regression, logistic regression on the encoded item, and boosting on it, the first rung ranges +0.173 to +0.184, the second +0.091 to +0.104, and the reduction 42% to 48%. Encoding a *shuffled* item column under the same pipeline returns

-0.0001 ± 0.0016 and $+0.0042 \pm 0.0020$ on the group-aware rung. The first collapses to zero; the second does not, and it is eleven standard errors from zero, so it is a real systematic effect rather than noise and it must be bounded on both rungs and not one. Run on the intake rung as well, the same control returns $+0.0002 \pm 0.0015$ and -0.0036 ± 0.0025 . Subtracting each rung’s own residual from that rung’s gain moves the reduction from 42.7% to 42.6% under the encoded logistic model and from 47.1% to 50.5% under boosting. The correction is therefore worth at most 3.4 percentage points and, where it bites, it makes the reduction larger.

Three target definitions.. Because reassignment is common, we repeat the ladder at stricter thresholds. Requiring two or more reassignments (21.8% of incidents) gives $+0.131$ and $+0.068$; three or more (10.6%) gives $+0.151$ and $+0.080$. The magnitude is threshold-dependent and not monotone, but the interval on each +group rung excludes zero (we compute intervals for that rung only), the ordering never reverses, and the reduction stays between 43% and 49%.

Does it survive a change of task?. A gap on one target may be a property of it, so we repeated the ladder on two further outcomes within the primary organisation. *Reopening* is close to an independent failure mode: 2,096 incidents, correlating with reassignment at $+0.14$. *Long handling* — handle time above the training third quartile — correlates at $+0.40$, so it is corroboration only.

On reopening, item identity is worth $+0.083$ against the intake block and $+0.055$ once the group is admitted, a reduction of 33%; on long handling, $+0.118$ and $+0.078$, a reduction of 34%. Every rung sits outside its matched-dimension null. But the reduction is what we argue transfers, so it needs its own interval, and paired bootstraps give $[40, 48]$ on reassignment, $[28, 38]$ on long handling and $[-1, 60]$ on reopening. On the one target we argue is near-independent the reduction is *not* resolvably different from zero ($P \leq 0$ is 0.03), and long handling, correlated at $+0.40$ with the primary target, cannot carry a replication alone. Reopening is rare and its gains stand only 5.5 and 4.2 pooled standard deviations from their nulls.

Cross-target replication within one organisation therefore fails to resolve, while cross-organisation replication succeeds. We take the second as the stronger evidence, because the objection it answers — that the effect is a property of one estate — is the one a reader should press hardest, and because

both Volvo intervals exclude zero. The reduction is largest on the target the opening group is most directly about, which is consistent with Section 7 rather than predicted by it: the mechanism says nothing about how the reduction should order across targets.

7. Where the Difference Goes

The two rows of Table 2 differ by 0.080, which is not itself informative: for any two variables, the drop in one’s gain when the other joins the baseline equals the drop in the other’s. Read four ways, the item is worth +0.183 over intake and +0.103 once the group is present; the group is worth +0.082 over intake and +0.002 once the item is present.

This section is weaker than the version we published previously, and the weakening is a finding rather than a concession. Of the four legs that supported the overlap reading, one has been withdrawn as an algebraic identity, one no longer resolves once its null is matched properly, and two survive.

What survives, first.. Adding the opening group to a model that already knows the item is worth +0.0017 [+0.0001, +0.0034]. It clears a matched-dimension null of -0.0009 ± 0.0005 that no draw reaches, so it is not the cost of adding columns; but across split points and penalties it ranges +0.0002 to +0.0072, so the honest statement is that its unique contribution is *under* 0.01 *AUC and not resolvable more finely*. Because the real column is nearly constant within an item while the null’s is drawn per row, this null is matched on dimension but not on that collinearity, and a sceptical reader should treat the interval as the weaker of the two claims.

What survives, second.. Item identity does not determine the group, but predicts it far better than chance: a modal-group lookup on item identity reproduces which group logged the incident on 90.3% of test incidents against 78.6% for always guessing the largest, while class-balanced accuracy reaches only 34.1%. Item identity resolves the coarse contrast — central desk or specialist team — and much less of the fine one. The two variables share 1.47 bits of mutual information.

A margin that was a knob.. Randomising the group label *within* each item — which destroys the label’s own identity while preserving what the item implies about it — retains 91% of the group’s +0.082 gain. We previously compared that against a floor built by randomising within a random partition *of items*

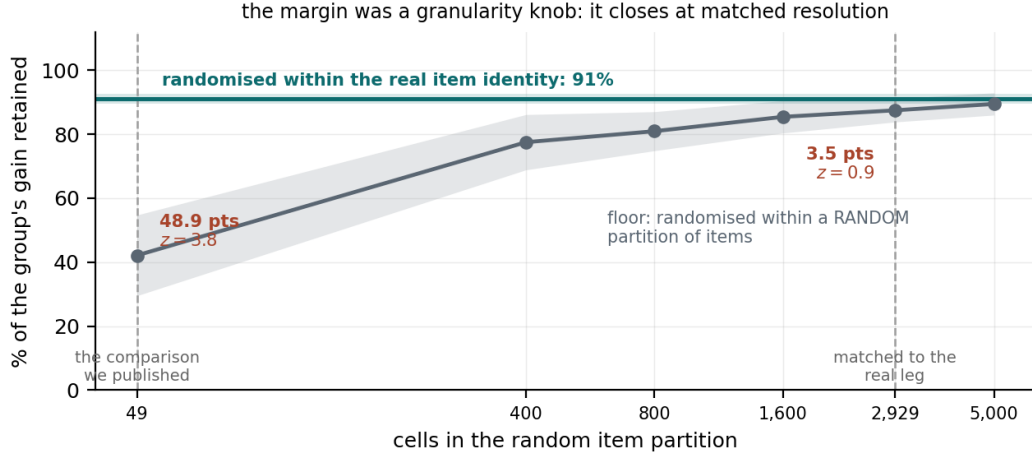


Figure 4: Randomising the group label within each item, against the same randomisation within a random partition of items, as a function of how fine the partition is. The margin we published was measured at 49 cells; the leg it bounds partitions on 2,929. It closes as the partition approaches the granularity of the real leg.

at 49 cells, matching the opening group’s own cardinality, which retains 42%, and reported the 49-point margin as evidence that item identity specifically was doing the work.

That comparison is not matched. The real leg partitions on item identity, which is 2,929 cells; the floor was drawn at 49. Retention rises monotonically with the fineness of the partition, so the margin is a function of a knob that was set to the value maximising it. The stated ground for stopping the sweep — that at one cell per item the null becomes the real thing — is wrong, because a *random* partition of n items into n cells is not the identity partition and stays routing-blind. Swept out to matched granularity the floor retains $87.5\% \pm 3.6\%$ against the real leg’s $91.0\% \pm 1.6\%$: a margin of 3.5 points at $z = 0.9$, against this project’s own resolvability bar of $|z| > 3$. We therefore withdraw the margin. What the sweep shows is that retention is high at fine granularity whether or not the cells mean anything, which is a fact about granularity and not about routing.

A withdrawn measurement: the asymmetry was an identity.. We previously reported that, without any estimator, item identity carries 60.4% of the opening group’s information while the group carries 19.6% of the item’s, and read the asymmetry as evidence that the mechanism runs from item to group. The reading is unsupported and the measurement is circular. The uncertainty co-

efficient $U(A | B) = I(A; B)/H(A)$, and mutual information is symmetric, so the ratio of the two coefficients is identically $H(\text{item})/H(\text{group})$ — a function of two marginal entropies and of nothing else. On this log that ratio is 3.09, and the two coefficients, their shuffled floors, and the floor-corrected excesses we published all stand in that same ratio to six decimal places. Any two variables whose entropies stand in that ratio produce those two numbers, whether or not either determines the other. We withdraw the asymmetry and the directional claim it supported. The mutual information itself, 1.47 bits, is a real measurement and is reported above; its *direction* is not measurable this way.

The reduction is graded in the field's resolution.. Coarsening the group and re-running the ladder gives reductions of 27%, 28%, 36% and 44% at two, four, eleven and 49 levels. This is a consistency check rather than an independent prediction, because at each resolution the reduction is close to that coarsened field's own gain over intake — by the identity above, the same statement as the first measurement — and the two coarsest points have overlapping intervals, so only the trend is claimed. A single binary flag — central desk or not — already recovers 61% of the group's baseline gain and 63% of the reduction. The field that must be admitted is close to one bit, far cheaper to reproduce elsewhere than a 49-way taxonomy.

Is the field really free? One objection goes to the heart of the reading. The opening group may be free only because a human at the desk already knew what the ticket was about — in which case it is undocumented configuration knowledge, and a baseline containing it is not a CMDB-free baseline. A third reading is at least as likely and we had missed it: the contrast may be an intake-channel indicator, separating an out-of-hours or overflow or automated route from the central desk, in which case it is neither independent of the item nor configuration knowledge held by a person. The one-bit result makes the question concrete without settling it: the contrast doing most of the work is the central desk against everything else, and Section 4 declines to say what everything else is. The measurement is the same under all three readings; the moral is not. If the group is independent, a CMDB is worth half what the naive comparison says. If it is tacit configuration knowledge, our +0.103 is a lower bound and the organisation already holds much of what the CMDB would tell it, unpaid for. The log cannot identify which is true, so we choose none of them. What survives either way is that the business case cannot be written against the intake block alone.

What we do not claim.. The reverse framing — that the item column “stands in for” the group — is not established. We measured it, by randomising items within groups, and obtained 44% of the item’s gain — but a routing-blind equal-size partition of items retains *more*, 56%, and one whose cell sizes follow the group’s own mass profile retains 25%. The leg is floor-dominated and we exclude it. Nor do we claim a direction of causation: the log records no counterfactual, so whether the item drives who logs the ticket or both follow a third cause is not identified here. Our claim is about what an analyst’s baseline choice does to a measured number, which does not depend on which is true.

Two fields we did not admit, and why.. Our baseline is not a terminus either. Hour of day and day of week are free and creation-time; adding both moves the item’s value from +0.103 to +0.099. **Service Component WBS (aff)** is harder: 100% populated, free, and admitting it takes the measured value to +0.023. We exclude it as near-deterministic in the item — only 58 of 2,929 items carry more than one value — so it is configuration data under another name. That criterion cuts both ways: 2,060 of 2,554 training items also map to a single opening group, though those items carry just 8.8% of incidents, so the high-volume ones are the multi-group ones. We know of no principled threshold that admits the group and excludes the service component. A reader who draws the line differently should read our headline as +0.023 — which is the decision this paper is about, and why we report a ladder rather than a number.

8. What the Difference Buys

An AUC difference is not a quantity a service desk can act on, so earlier versions of this paper stated the same comparison as detection at a fixed review capacity: rank the test incidents by predicted risk, take the top 5%, and count how many reassignment-bound incidents each model surfaces. That framing is withdrawn here, and the reason is the second of this round’s two corrections.

8.1. Why the capacity framing was withdrawn

The paper already disclosed that the naive baseline cannot rank finely: four low-cardinality intake fields, one of them the redundant Priority column, emit only 23 distinct scores over 13,637 test incidents, so at 5% capacity just

Tie-breaking policy	naive extra	honest extra	factor
random — the only implementable one	+271	+63	4.3
oracle: positives first in each tie	−26	+24	—
adversarial: negatives first	+608	+104	5.8

Table 3: Extra reassignment-bound incidents surfaced at a 5% review capacity by adding item identity, against a baseline that includes the opening group (honest) and one that omits it (naive), under three policies for ordering rows *inside* a tied block. The oracle and adversarial policies use the outcome and are unavailable to an analyst; together they bracket what tie-breaking alone can do. Medians of 400 paired bootstrap draws over test rows.

47 rows sit strictly above the cut and the rest come from a tied block of 1,944. It said the factor was therefore partly about rank resolution rather than information, that the two had not been separated, and moved on. Separating them destroys the number.

The naive top-5% is a draw from a tie.. Of the 682 incidents the naive baseline nominates, 635 — 93.1% of the review budget — come from a single tied block, and that block is reassigned at 0.534 against 0.372 overall. How many of the right incidents the draw yields is therefore fixed by the block’s own base rate and by nothing the model knows. The 47 rows the model does rank strictly above its cut contain 9 reassignment-bound incidents, a rate below the base rate: at the top of its own ranking the naive baseline is worse than guessing. The group-aware baseline is in a different position, with 590 of 682 ranked strictly above the cut and only 92 drawn from a block of 246.

The factor’s sign is set by the tie-break.. Table 3 prices that. Ordering rows inside the tied block optimally — which uses the outcome, so no analyst can — reverses the sign of the naive contrast: item identity surfaces 26 *fewer* incidents than the intake block alone, not 271 more. Ordering them adversarially gives +608. Nothing any model knows changes between those three rows. A quantity whose sign is set by how a coin lands inside a 1,944-row block is not a measurement of information, and we withdraw it.

A reader may object that the oracle policy favours the coarser arm by construction: it has the larger block to reorder, so it has the most to gain from an ordering it cannot obtain. That is the objection restated as the finding. A contrast whose magnitude is governed by the size of one arm’s tie block, rather than by what either arm knows, is measuring the block. The

random policy is the only implementable one and its value is the one an analyst would see; but its value is the expected yield of a draw from a block whose composition is a property of the cohort, not of the ranking, and the paper should not have read a difference in information off it.

The obvious repair is impossible.. It is natural to answer the objection by giving the naive baseline a finer representation, and we tried: cross-fitted target encoding of the intake block, per field and as a single composite key, on the same machinery used for the item column in Section 6. It cannot work. The four intake fields take 23 distinct combinations, so every row sharing a combination shares a score under *any* function of those fields, and combinations unseen in training collapse onto the training prior — the composite encoding emits 19 distinct scores, fewer than the one-hot model’s, not more. The tie block is not an artifact of our estimator. It is what having four low-cardinality intake fields is.

What survives is narrower and we state it as such: the honest arm’s extra catches are positive under every tie policy, including the oracle (+24 [+9, +40]), so knowing the affected item does surface more reassignment-bound incidents at this capacity than the group-aware baseline does. What does not survive is the ratio, and with it the claim that omitting the free field overstates the operational gain by more than it overstates the AUC gain.

8.2. Net benefit, which needs no capacity

Vickers and Elkin [26] supply the instrument the question actually needs. Fix a threshold probability p_t — the risk at which a desk judges a review worth doing — act on every incident scored at or above it, and compute

$$\text{NB}(p_t) = \frac{\text{TP}}{n} - \frac{\text{FP}}{n} \cdot \frac{p_t}{1 - p_t}.$$

The weight $p_t/(1 - p_t)$ is the exchange rate the threshold implies: at $p_t = 0.25$ one missed reassignment-bound incident is worth three needless reviews. Net benefit is in true positives per incident, so we report $1000 \times \text{NB}$, which reads as reassignment-bound incidents caught per thousand arrivals net of the reviews spent catching them. It assumes no capacity, it is defined across the whole range of exchange rates a reader might hold rather than at one we picked, and — the point that matters after Section 8.1 — a threshold admits or excludes a whole tied block, so no tie is ever broken.

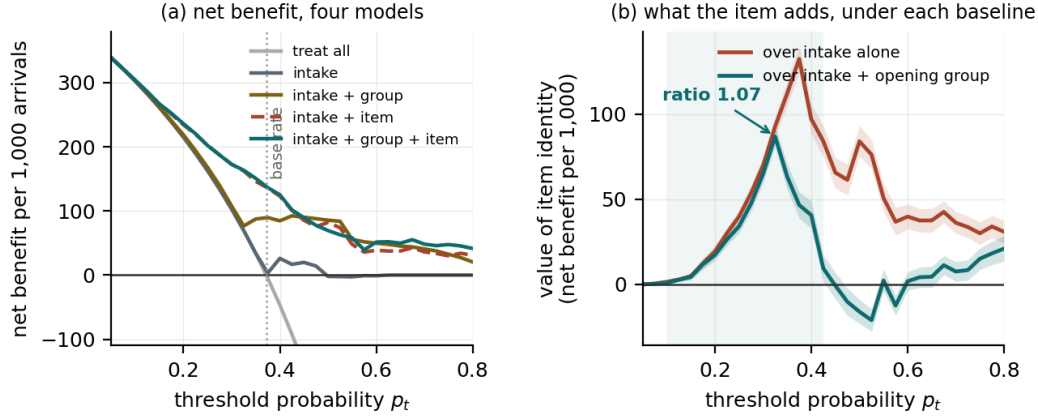


Figure 5: (a) Net benefit per 1,000 arrivals for the four models and for treating every arrival, across threshold probabilities. (b) What item identity adds under each baseline, with 2,000-draw paired bootstrap bands. The shaded region is where the group-aware increment’s interval excludes zero. The two curves are indistinguishable until the threshold passes this cohort’s base rate.

The scores are probabilities enough for this.. Net benefit is meaningless for a model whose scores are not calibrated, and a one-hot logistic model with 2,554 sparse columns at a fixed penalty is not obviously calibrated. Brier scores run 0.232 for the intake block, 0.206 with the opening group and 0.189 with the item; calibration slopes run 1.391, 1.185 and 1.040, so the intake block is the poorly calibrated arm and the item-aware model is close to ideal. Because a slope away from one moves net benefit, we repeat the whole analysis on scores recalibrated on a held-out tail of the training half — the model refitted on the first 85% of training, scored on the last 15%, and the scaling applied to test, with no test outcome entering it. No conclusion below changes.

Where the item pays, and by how much less than it appears to.. Over a grid of 31 thresholds from 0.05 to 0.80, the group-aware increment is positive with its interval excluding zero at 20 points, in a contiguous run from 0.100 to 0.425 — the region either side of this cohort’s base rate. The naive increment is positive and resolved at 29 of the 31. At the threshold where item identity is worth most, $p_t = 0.325$, it adds 86.9 reassignment-bound incidents per thousand arrivals over the group-aware baseline and 92.8 over the intake block: omitting the free field overstates its value there by a factor of 1.07. This is the number that replaces the withdrawn 4.3, and it is smaller than

p_t	over intake + group	over intake alone	ratio
0.20	+17.2 [14.0, 20.4]	+19.3 [16.0, 22.5]	1.12
0.30	+64.9 [59.8, 70.2]	+69.9 [64.7, 74.8]	1.08
0.40	+40.7 [33.0, 48.9]	+97.2 [88.6, 106.0]	2.39
0.50	-16.1 [-23.0, -8.9]	+84.1 [75.5, 93.2]	—
0.60	+1.8 [-3.7, 7.2]	+39.8 [32.5, 47.2]	—

Table 4: What item identity adds, as net benefit per 1,000 arrivals, under the two baselines. A ratio is not quoted where the group-aware increment’s interval includes zero or lies below it, because a ratio whose denominator is crossing zero is not a quantity.

the ratio of 1.8 between the same two gains measured as AUC.

And where it does not pay at all.. Above the resolved region the increment decays and then turns negative: at four grid points between 0.475 and 0.575 the group-aware increment’s interval lies entirely below zero, reaching -16.1 $[-23.0, -8.9]$ per thousand at $p_t = 0.50$. At a one-for-one exchange rate, adding item identity to a model that already knows the opening group makes the desk worse off on this task. We have no mechanism to offer for that beyond the obvious one — a model with 2,554 item indicators can be confidently wrong where a coarse model abstains — and we report it because it is the operating region in which a desk that reviews a small share of arrivals is working, and because it is the honest reason the capacity framing produced the number it did. The earlier version measured the overstatement at the one operating point where its own group-aware arm had the least to give.

8.3. What the operational statement does not say

The threshold ladder of Section 6 is the operational statement of the same robustness, at the three target definitions reported there. None of this says a surfaced incident is a corrected one, and net benefit does not claim otherwise: it counts detections against reviews at a stated exchange rate, and what a detection is worth to a particular desk is a quantity this log cannot supply.

9. One Thing We Could Not Establish

A third field, the knowledge-article reference, sits on the same `Open` row, is 100% populated, and costs nothing. Adding it to the baseline raises AUC to 0.805 and drives the measured value of item identity to -0.003 . Whether

that figure belongs in Table 2 depends on whether the field is available at creation, and we could not determine this.

Two structural tests both fail. A field that varies within an incident is certainly a per-event observation; the knowledge reference never varies — but neither does **Interaction ID**, which has 45,426 distinct values for 45,455 incidents and is plainly a creation-time key, so constancy is evidence of granularity, not timing. Exact identity with the closed record fares no better: the knowledge reference matches its closed-record column for 100.000000% of incidents, but so does **Interaction ID** for 99.997628% of the 42,151 single-interaction incidents — one mismatch against a pass mark of 100.000000%.

We therefore make no claim about this field. The -0.003 is also inside the dimensionality null of Section 6 and changes sign under penalty tuning, so it is not resolvable on either ground. This is the paper’s thesis turned on itself, and that reading is correct: our headline is contingent on a field-admission decision we cannot adjudicate. The transferable part is the negative — a field being free, populated and present on an opening record does not make it creation-time. The collection ships an interaction detail file we did not obtain; if it records this field at creation the question is answerable, and our inability to settle it is a limit of the three files we hold, not of the export.

10. Limitations

Estate and era.. Two organisations, two platforms, both on-premises products, and six months of 2013 and 2014 data in the primary case. The values $+0.183$ and $+0.103$ are properties of one estate; what we argue transfers is that the gap exists and is large. Practice has moved since: discovery tooling auto-populates a large fraction of a modern estate, event-driven incidents arrive with the item already stamped by the monitoring source, and service mapping rather than item identity is where much current CMDB spend goes. The counterfactual we model — an organisation with no item field at all — is not the counterfactual a 2026 buyer faces, which is about relationship depth and data health rather than existence. Section 5 is the part of this paper that speaks to that buyer, and it says the service layer carries most of the measured value.

Data quality.. Our item field is 100% populated because the tool copies it from the originating interaction record. Real CMDBs are not this clean, and a lower population or accuracy rate would reduce the item’s measured

value. Our +0.103 should be read as an upper bound for an estate whose configuration data is worse than this one’s, which is most of them.

Intake channel.. The free field’s informativeness depends on an intake mix in which two thirds of tickets pass a central desk. Where tickets are opened by end users through a portal or by an integration account, the field may carry far less, or may not exist in a comparable form. The one-bit result implies the contrast that matters is central desk against everything else, and that contrast is a property of how the organisation takes work in.

Free text.. Neither log carries a usable short description, and every real deployment has one. Kapel et al. [11] rank two free-text fields above CI Name for a related target. If free text absorbs much of the item’s value, the field-admission argument becomes stronger and the CMDDB’s measured value smaller; we cannot test it here, and it is the most important open question this study leaves.

Construct and mechanism.. Reassignment is a proxy for misrouting and also fires on legitimate escalation; the within-item shuffle preserves item membership but not other correlates of identity, so it bounds the proxy share rather than isolating it; and the opening group’s distribution drifts between our halves. After this round’s withdrawals, we do not claim a measured direction for the mechanism.

The operational statement is a detection statement.. Net benefit counts detections against reviews at a stated exchange rate. It does not say that a detected incident is a corrected one, that a desk given the warning would route differently, or what a correct routing is worth. Section 8 also reports a region of thresholds in which the item is worth nothing or less than nothing over the group-aware baseline, and we have no mechanism for that beyond over-confident predictions from a high-dimensional model.

11. Corrections

Eight errors of our own are reported as results rather than edited away, because each is an instance of the failure this paper is about — a control, a claim or a baseline asserted rather than checked — and because the pattern across them is more informative than any one.

1. **A null that could not fail.** The mechanism’s floor was originally drawn by assigning cell labels per *row*, so two incidents sharing an item landed in different cells and the association was destroyed by construction. The null could only return zero, and we published a margin of 89 points against it (Section 7).
2. **The same null, rebuilt, was still a knob.** Rebuilt at item level it was swept only to 800 cells while the leg it bounds uses 2,929, and the reported margin was taken at the cell count that maximised it. At matched granularity the margin is 3.5 points at $z = 0.9$ and we now withdraw it. This is the fifth time a null in this project has been drawn at the wrong level.
3. **An asymmetry that was an algebraic identity.** The estimator-free evidence for the direction of the mechanism was a ratio of two uncertainty coefficients, which is identically the ratio of two marginal entropies. It was published for eight rounds with both entropies sitting in the same results file, one division away.
4. **A factor printed without an interval.** The capacity-table factor was first given as a point estimate, on an unmatched pair of models, in a paper that elsewhere refuses to resolve +0.0017 past two decimals.
5. **A central field whose meaning we asserted.** The opening group was described as the routing queue for several versions. It is the group that logged the incident (Section 4).
6. **A claim about a public dataset that our own repository refuted.** We wrote that the third public ITSM log has no affected-item field. It has one, populated on every case, and a script in our own repository had been treating it as such (Section 4.2). This was the sole support for describing the study as single-organisation of necessity, which we had presented as a constraint rather than a choice. It was a choice.
7. **An operational factor whose sign was a tie-break.** Correction four gave that factor an interval; this round asked what it measured. At a 5% review capacity 93.1% of what the naive baseline nominates is drawn from one tied block, and reordering rows inside that block — which changes nothing any model knows — moves the naive arm from -26 to $+608$. We withdraw the factor and rebuild Section 8 on net benefit, which never breaks a tie. The replacement number is 1.07, not 4.3 (Section 8.1).

8. **A control run on one of the two rungs it bounds.** The shuffled-item encoder null was run on the group-aware rung only, while the reduction it is meant to bound is computed from two rungs, and its residual under boosting is eleven standard errors from zero rather than noise. Run on both rungs, the correction moves the boosting reduction by 3.4 percentage points, and upward (Section 6).

Six of the eight flattered the result, one — the dataset claim — excused its principal limitation, and one, the eighth, would have made the result larger had we noticed it earlier. We draw the obvious inference about the direction in which unchecked assertions travel, and note that the paper’s verification apparatus, which recomputes every number it prints, would not have caught any of them: all eight are claims about what a number *means*.

12. Conclusion

What configuration data contributes to this task is a stable identifier under which outcome history accumulates, not the configuration attributes a CMDDB is usually bought for: a per-item outcome rate with no model reaches 0.744 against the full model’s 0.748, and a 256-way service-component grouping captures three quarters of what instance-level identity is worth, with instance identity adding +0.023 over it. The layer an organisation is buying, and not the fact of buying, is what the measured value turns on.

The size of that measured value is set by a second choice. Knowing which configuration item an incident concerns is worth +0.183 or +0.103 AUC for predicting reassignment, depending on whether the baseline includes one field the organisation already records: which group logged the ticket. The reduction runs 36.1% to 48.3% across the design space, it survives the admission of free congestion features, and the pattern replicates on a second organisation. The difference is an overlap — a model given the item has already been told most of what the group would say — so a business case built on the first figure is measuring a CMDDB plus a field the organisation was already recording for nothing.

Three qualifications belong in the same breath. Which figure is right depends on an admission line we cannot draw principledly: admit the service component instead and the number is +0.023. Stated operationally rather than as AUC, the overstatement is far smaller than we previously reported — 1.07 at the threshold where the item is worth most — and we withdraw

the factor of 4.3 that an earlier version printed. And in a band of thresholds above the base rate, the item is worth nothing at all over a group-aware baseline. The transferable claim is not a number. It is that the number is chosen, and that whoever writes the business case is choosing it.

Acknowledgements and Reproducibility

The manuscript was subjected to sixteen rounds of adversarial review, the later rounds machine-assisted, in which each round was asked to break the previous version’s claims; the corrections in Section 11 came from that process. The author set every research question, adjudicated every proposed correction against the data, and is responsible for the content. The datasets are public [8, 22]; code, figures and a checker that recomputes every quantity here: <https://github.com/sudhirdixit1/cmd-b-field-admission>.

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